The Effect of Anthropomorphism on Investment Decision-Making with Robo-Advisor Chatbots

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THE EFFECT OF ANTHROPOMORPHISM ON INVESTMENT DECISION-MAKING WITH ROBO-ADVISOR CHATBOTS

Research paper

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Abstract

Robo-advisors promise to offer professional financial advice to private households at low cost. However, the automation of financial advisory processes to fully-digitalized services often goes hand in hand with a loss of human contact and lack of trust. Especially in an investment context, customers demand a human involved to ensure the trustworthiness of digitalized financial services. In this paper, we report the findings of a lab experiment (N=183) investigating how different levels of anthropomorphic design of a robo-advisor chatbot might compensate for the lack of human involvement and positively affect users' trusting beliefs and likeliness to follow its recommendations. We found that the anthropomorphic design influences users' perceived social presence of the chatbot. While trusting beliefs in the chatbot enhance users' likeliness to follow the recommendation by the chatbot, we do not find evidence for a direct effect of social presence on likeliness to follow its recommendation. However, social presence has a positive indirect effect on likeliness to follow the recommendation via trusting beliefs. Our research contributes to the body of knowledge on the design of robo-advisors and provides more insight into the factors that influence users' investment decisions when interacting with a robo-advisor chatbot.

Keywords

Robo-advisory; chatbot; anthropomorphism; experiment; social presence; trusting beliefs

1 Introduction

Robo-advisors are web-based systems that have received much attention in recent years because of their potential to offer professional financial advice to private households at low-cost (Fisch et al. 2018; Jung et al. 2019; Jung, Dorner, Glaser, et al. 2018). They use different approaches to digitalize and automate wealth management (Ludden et al. 2015; Sironi 2016), based on state-of-the-art algorithms in portfolio-management and are hence detached from emotionally biased decisions (Tertilt and Scholz 2017). However, the automation of the financial advisory process to a full-digitalized service, while maintaining characteristics like trust or service quality, is a challenging task (Hodge et al. 2018; Jung, Dorner, Weinhardt, et al. 2018). In a recent study by Jung et al. (2018), users requested to first have a personal conversation with a human advisor before making an investment with the robo-advisor. Interestingly, their motivation was not to receive more information, but primarily to convince themselves of the robo-advisor's trustworthiness by double-checking with a human advisor. Accordingly, it seems like personal contact, via the human advisor, plays an important role in users' confidence-building and acceptance of robo-advisors' investment advice. However, it is not clear whether a human advisor is actually required or whether a more human-like or anthropomorphic design of the robo-advisors might compensate for the lack of human contact and have positive effects on the users' trusting beliefs and acceptance of the robo-advisors' investment recommendations.

While digitalization improves efficiency, it removes the social aspect of human-to-human interaction by replacing it with human-to-computer interaction (HCI). However, research suggests that HCI is fundamentally social because certain design features of a computer can trigger emotional, cognitive, and behavioral reactions by users that are appropriate when directed at other humans, but inappropriate when directed at computers (Nass et al. 1994; Nass and Moon 2000). These reactions can be explained by the "Computers are Social Actors" (CASA) paradigm and social response theory. It posits that humans tend to form similar expectations about interactions with computers when they show human-like design features as they do with humans (Nass and Moon 2000). For example, several studies have shown that social aspects play an essential role in establishing trustworthiness and believability of advisory agents by increasing the agents' social presence (Nass et al. 1994; Sproull et al. 1996). In fact, several studies about recommendation agents show that social presence enhance users' trusting beliefs and ultimately usage intentions (e.g., Hess et al. 2009; Lee and Choi 2017; Qiu and Benbasat 2009). Thus, increasing the social presence of robo-advisors using an anthropomorphic design might positively affect users' trusting beliefs and their likeliness to accept recommendations by robo-advisors.

However, although the design of robo-advisory has been increasingly investigated in the last years (e.g. Ivanov et al. 2018; Kobets et al. 2018; Tertilt and Scholz 2018), little attention has been paid to the effect of anthropomorphizing the robo-advisor itself. Since robo-advisors are intended to replace human advisors in financial advisory, the question arises of how "human-like" a robo-advisor has to be in order to provide convincing investment recommendations. Moreover, it is unclear whether a certain level of anthropomorphism (i.e., a high level of anthropomorphism meaning a high level of human-likeliness) more positively affects the users' acceptance of the robo-advisor as a trustworthy financial advisor and ultimately following its recommendation, than other levels. In the current body of knowledge, the role of the anthropomorphizing financial robo-advisory is not well understood, and only little evidence is available (e.g. Hodge et al. 2018). Conversational user interfaces and chatbots are becoming increasingly popular (Følstad and Brandtzæg 2017). Due to their use of natural language, they promise to offer a more natural and human-like interface to digitalized services. Consequently, one way to anthropomorphize robo-advisors is to use a conversational user interface (CUI) (McTear 2017) and implement the robo-advisor as a text-based conversational agent (i.e., a chatbot). Robo-advisor chatbots (RACs) have received little attention in research (e.g. Day et al. 2018) and there is a need to better understand the outcomes of interacting with RACs on the users' trusting beliefs and likeliness to follow their investment recommendations. In our research, we investigate whether different levels of anthropomorphism of the RAC affects users' perception of social presence, trusting beliefs, and ultimately their likeliness to follow the RAC's advice. Thus, we address the following research question:

How do different levels of anthropomorphic design of a robo-advisor chatbot influence the users' perceived social presence, trusting beliefs, and their likeliness to follow its advice?

In this paper, we report the findings of a between-subjects laboratory experiment that investigated the effect of anthropomorphized RACs in the context of investment decision-making as an important specific task in financial decision-making. Drawing on social response theory (Nass and Moon 2000), we investigate the effect of three different levels of anthropomorphism of a RAC on the users' perceptions of the RAC and their investment decision-making. We found a significant effect of a higher level of anthropomorphism on the users' perception of the RAC's social presence. While trusting beliefs in the chatbot enhance users' likeliness to follow the recommendation by the chatbot, we do not find evidence for a direct effect of social presence on likeliness to follow its recommendation. However, social presence has a positive indirect effect on likeliness to follow the recommendation via trusting beliefs. Our research contributes to the body of knowledge about the design of robo-advisory by emphasizing the importance of purposefully designing RAC and confirming the importance of users' trusting beliefs on their investment decision-making.

2 Theoretical Background and Related Work

Robo-advisors are systems providing financial advice to investors based on algorithms that analyze financial information without human intervention (Hodge et al. 2018). However, the interaction with a robo-advisor is similar to financial advisor process with a human: The robo-advisor asks questions about the individuals' financial risk-preferences, their long- and short-term investment goals, and subsequently gives an investment recommendation that fits the users' needs (Hodge et al. 2018; Jung, Dorner, Weinhardt, et al. 2018). Research on the design of the interaction between users and robo-advisors is scarce as the interest in this technology from an academic point of view just started a couple of years ago. Existing research focuses primarily on the design of the robo-advisor process and the digitalization of the financial advisory process in general (Jung, Dorner, Glaser, et al. 2018). Kobets et al. (2018) investigate the decision-making process of robo-advisor users and provide a general framework for providing user-specific decision support with robo-advisors. Tertilt and Scholz (2018) investigate the capabilities of robo-advisors to provide user-sensitive risk-assessment. Furthermore, Ivanov et al. (2018) propose a guideline to implement risk-assessment in robo-advisory based on user-specific input. So far, research has primarily focused on how the financial advisory process can be digitized using robo-advisors. However, there is evidence that suggests that many investors still prefer a human advisor (Jung, Dorner, Weinhardt, et al. 2018). Previous research in the context of HCI has shown that increasing the anthropomorphism of a system may make up for a lack of human contact by increasing the perceived social presence of the system (e.g., Araujo 2018; Hess et al. 2009; Qiu and Benbasat 2009). Thus, investigating the effect of anthropomorphizing a robo-advisor chatbot might provide an interesting avenue for research and contribute to the knowledge base on the design of robo-advisory.

2.1 Anthropomorphism and Social Presence

Anthropomorphism is the attribution of "*humanlike properties, characteristics, or mental states to real or imagined nonhuman agents and objects*" (Epley et al. 2007, p. 865). In essence, it represents a human heuristic that helps to understand unknown agents by applying anthropocentric knowledge (Griffin and Tversky 1992). Research has found that it can be applied to all nonhuman agents such as nonhuman animals, technological devices, or mechanical devices and the extent to which individuals engage in anthropomorphism depends on different factors (Epley et al. 2008). For example, children tend to anthropomorphize more often than adults (Carey 1985). More importantly, the tendency to anthropomorphize an object depends on the presence of characteristics expressing a degree of humanness (Epley et al. 2007). Therefore, the more specific anthropomorphic features/cues are embedded in the design of an information system (IS), the more likely human users are to anthropomorphize it (Pfeuffer et al. 2019). Extant research shows the important role of anthropomorphic cues in influencing users' perceptions of IS and behaviors when interacting with an anthropomorphic IS. Moreover, research has shown that anthropomorphic cues influences users' perceptions of social presence (Araujo 2018; Go and Sundar 2019; Kim and Sundar 2012). In the context of HCI, the concept of social presence broadly refers to the feeling of warmth, sociability, and human contact during the interaction that can be created without actual human contact (Gefen and Straub 2004). Social presence has been identified as a key factor in the design of IS in online settings (Hess et al. 2009) and as an important driver of users' trusting beliefs (Cyr et al. 2007; Hess et al. 2009; Qiu and Benbasat 2009). For example, Lu, Fan, and Zhou (2016) showed that social presence strongly influences consumers' online shopping behavior. Moreover, other studies suggest that social presence (e.g., induced by an avatar) can increase users' satisfaction and trust in an online shopping context (Etemad-Sajadi 2016; Holzwarth et al. 2006). While previous IS research has provided broad evidence of the effect of anthropomorphic features/cues, there is a lack of understanding on how these features influence users' investment decisions. This is important because a higher level of human-likeliness of the robo-advisor might have a positive effect on the users' trusting beliefs and the likeliness to follow the recommendation by the robo-advisor. However, there is only little research available. Hodge, Mendoza, and Sinha (2018) examine the effect of humanizing robo-advisors in contrast to human advisors in the context of investor judgements. They conducted an experiment and manipulated the type of advisor (human vs. robo-advisor) and the level of humanization (low vs. high). They found that investors are more likely to follow the recommendation of a robo-advisor with a low level of humanization. In contrast, investors are more likely to follow human advisors exhibiting a high level of human characteristics. Hence, there is more research required to investigate the effect of anthropomorphizing a robo-advisor. Recently, chatbots have also been introduced as financial advisors (Day et al. 2018) and using a CUI might provide new insights into how the trusting beliefs towards a robo-advisor could be increased in order to increase users' likeliness to follow the robo-advisor's investment recommendation.

2.2 Chatbots and Conversational Agents

In previous research, many terms have been used to describe IS that allow users to interact with them using natural language (e.g., conversational agent, chatbot, or digital assistant) (Gnewuch et al. 2017; Maedche et al. 2019). In IS research, the most commonly used term for this class of systems is "conversational agent", which includes embodied (Cassell et al. 2000; Derrick et al. 2011), voice-based (e.g., Amazon's Alexa), and text-based conversational agents (i.e., chatbots) (Araujo 2018; Dale 2016; Gnewuch et al. 2017). Since chatbots are often found in messaging applications and on websites, users interact with them using text-based natural language (Følstad and Brandtzæg 2017). The first chatbot ELIZA, developed in the early 1960s, was able to simulate a human conversation based on pattern-matching algorithms (Weizenbaum 1966). Much research has been conducted since then to improve existing algorithms for natural language processing and create new architectures (Sarikaya 2017). While most studies focus on their technical aspects, understanding conversational processes, and identifying elements that influence users' social perceptions of chatbots seems to be equally important (Følstad and Brandtzæg 2017; Jenkins et al. 2007). This is mainly because users interact with chatbots via natural language (i.e., a uniquely human quality) (Seeger et al. 2017). Moreover, chatbots are often implemented to fulfill the role of service employees (Larivière et al. 2017) and serve as recommendation agents to support consumers in their decision-making when searching for products (Qiu and Benbasat 2009).

Summed up, we identified a potential need for a more human-like design of robo-advisory (Hodge et al. 2018; Jung, Dorner, Weinhardt, et al. 2018). Moreover, there is only little research on the effects of using a CUI to interact with the robo-advisory (Day et al. 2018). In line with existing research, we assume that using a CUI and anthropomorphizing RACs will increase the perceived social presence, positively effects the users' trusting beliefs (Araujo 2018; Qiu and Benbasat 2009), and ultimately their likeliness to follow the recommendation by the RAC.

3 Research Model

To answer our research question and investigate the effects of anthropomorphizing RACs, we developed a research model and four hypotheses. The following section derives the hypotheses on the proposed effect of anthropomorphizing a RAC to the users' perceived social presence of the RAC (H1) and the interplay of users' perceived social presence, trusting beliefs, and likeliness to follow the RACs recommendation (H2-H4). Thus, our research model (see Figure 1) comprises our treatment as the independent variable *Anthropomorphism* and the dependent variables *Perceived Social Presence, Trusting Beliefs,* and *Likeliness to Follow Advice*. Furthermore, we control for different factors potentially influencing the likeliness to follow the recommendation by the RAC such as age, gender, disposition to trust in technology, and risk attitude. Besides, we control for common usage characteristics like users' experience with chatbots and experience with robo-advisors.

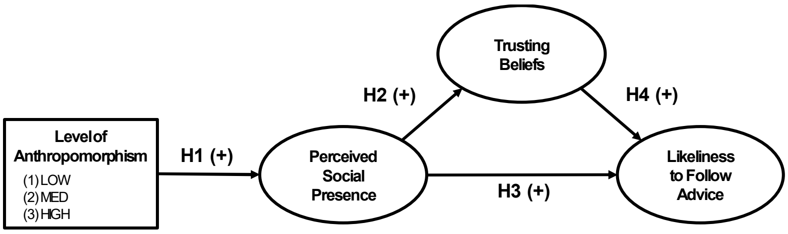

Figure 1. Research Model

There is substantial evidence that suggests a linear relationship between the degree of anthropomorphism in computers and users' perceptions of social presence (Epley et al. 2007; Gong 2008; Nass and Moon 2000). Nass and Moon (2000) state that "*the more computers present characteristics that are associated with humans, the more likely they are to elicit social behavior*" (Nass and Moon 2000, p. 97). While users to a certain degree respond socially to computers even without humanoid traits (Klein et al. 2002), this aspect is amplified by anthropomorphic cues such as more human-looking avatars (Nass et al. 1998). Anthropomorphic cues, such as a human-like avatar, greetings or politeness, make it more difficult for users to carefully elaborate the "ontological nature" of the RAC (Qiu and Benbasat 2009). This may lead them to evaluate, judge, and respond to the anthropomorphized RAC by subconsciously applying the same social rules they apply in everyday life (Nass et al. 1994, 1995; Qiu and Benbasat 2009). For example, the study of Qiu and Benbasat (2009) found that anthropomorphic cues, such as avatars, positively influence perceived social presence of a product recommendation agent. Therefore, users are expected to perceive the interaction with the anthropomorphized RAC as more interpersonal, the more anthropomorphic cues are present during the interaction. Hence, we assume that the different levels of anthropomorphizing the RAC will influence users' perceived social presence:

> **H1**: *An increased level of anthropomorphism of the robo-advisory chatbot is positively associated with users' perceived social presence of the robo-advisory chatbot.*

Trusting beliefs consists of three dimensions: competence, benevolence, and integrity (McKnight et al. 2002). Prior research found social presence as both an enabler and an antecedent of trust in the context business-to-consumer e-services provided by a website or by (recommender) agents (e.g., Gefen and Straub 2003; McKnight et al. 2002; Qiu and Benbasat 2009). More specifically, trusting beliefs can be formed by users' perceived quality of the information provided by the agent (Qiu and Benbasat 2009), such as the relevance of questions, the amount and scope of explanations, or how well recommendations conform to users' preferences. In addition, a higher level of social presence of the agent is found to be perceived as more transparent by the users, and thus, as more trustworthy (Hess et al. 2009). Especially during the first interactions with an agent, it is nearly impossible for users to accurately appraise the integrity and completeness of information provided by the agent. This so-called product-related evaluation is especially difficult for users with a limited product knowledge (Qiu and Benbasat 2009). Thus, one important source of trusting beliefs is the so called initial trust, which is formed upon "whatever information is available" (Meyerson et al. 1996; Qiu and Benbasat 2009). In face-to-face human interactions, trivial but important cues lead to the formation of perceived trustworthiness (Metzger et al. 2003). In the context of HCI, social presence (enabled through cues of the website or agent) is identified as such a non-product-related information that enables and antecedes trust (Gefen and Straub 2003; Lee and Nass 2004). Additionally, linguistic cues widely employed by human individuals increase users' perceived benevolence and credibility (Cassell and Bickmore 2000). Moreover, Hertzum et al. (2002) found lively personifications that lead to a higher social presence to be more convincing than agents with a less personified visual appearance. Summing up prior study results about the effects of perceived social presence on users' trusting beliefs lead to the following hypothesis:

> **H2**: *Users' perceived social presence of the robo-advisory chatbot is positively associated with users' trusting beliefs towards the robo-advisory chatbot.*

Robo-advisors are designed to help users in investment decision-making and require a trustworthy atmosphere (Jung, Dorner, Weinhardt, et al. 2018). Consequently, if users do not trust the robo-advisory agents, they are likely to reject their recommendations (Wang and Benbasat 2005). While the robo-advisory process has a lot of overlaps with the traditional financial advisory process, humans judgments about advisors can substantially differ between the virtual and the human agent (Hodge et al. 2018). Although prior research on HCI suggests that people perceive humans to be more likeable and social than computers, it remains unclear if this also makes humans more persuasive, especially when it comes to investment decisions (Burgoon et al. 2000; Fogg and Tseng 1999; Hodge et al. 2018; Nan et al. 2006). Many characteristics that lead to an increased persuasiveness in human-human interaction can be extended to situations in which humans interact with technology (Yoo and Gretzel 2011). Since social presence bridges the gap between

HCI and traditional human-human interaction (Nass and Moon 2000), it might mediate the effect between the RAC design (i.e. the anthropomorphic cues implemented in the RAC) and users' likeliness to follow the advice from the RAC. Additionally, research has shown that the implementation of social aspects that elicit social responses from users lead to a higher persuasiveness of the respective technology (Al-Natour et al. 2006; Wang and Benbasat 2005). Hence, we propose that an increased level of perceived social presence of the RAC will result in a higher probability of the users to follow the recommendation by the RAC:

> **H3***: Users that exhibit higher levels of social presence are more likely to follow the recommendation by the robo-advisory chatbot.*

When individuals are about to determine whether to follow a recommendation from someone else or not, the individuals' perception of the other person's competence and trustworthiness is of critical importance (van Doorn et al. 2017; Friestad and Wright 1994; Hodge et al. 2018; O'Keefe 2015). Trusting beliefs have been found to be effective mediators that influence trusting intentions (Lee and Turban 2001; McKnight et al. 2002; McKnight and Chervany 2001). While trusting beliefs describe users general impressions about the agent's trustworthiness, trusting intentions are "*intentions to engage in trust related behaviors*" (McKnight et al. 2002, p. 335). They are therefore related to actual actions, such as following a recommendation or not. Several other studies also found that perceived competence and trustworthiness are important dimensions when determining whether to follow a recommendation or not (van Doorn et al. 2017; Friestad and Wright 1994; O'Keefe 2015). These dimensions play a key role in mediating the relationship between the change in the likeliness of following the advisor's recommendation and the level of anthropomorphism (Hodge et al. 2018). Thus, summarizing the findings in literature, we propose the following hypothesis:

> **H4***: Users that exhibit higher levels of trusting beliefs are more likely to follow the recommendation by the robo-advisory chatbot.*

4 Method

To assess the proposed hypotheses and investigate the effects of anthropomorphizing a robo-advisor in the form of a chatbot, we conducted a controlled laboratory experiment. In line with existing research (Go and Sundar 2019; Gong 2008; Wünderlich and Paluch 2017), we decided to implement different levels of anthropomorphism in our artifact: low, medium, and high. In the experiment, the participants were asked to interact with a RAC, receive a recommendation for an investment, and decide for an investment in our fictitious scenario (i.e., follow the advice by the RAC or not). After this interaction, the participants filled out a questionnaire about their perceptions of the RAC and their interaction with it.

4.1 Participants

Our study was conducted in the behavioral research lab of a German University and we recruited the participants from a pool of university students. We decided to involve students in our study because we are interested in participants with only little prior knowledge of financial investments and usage of financial decision aids (such as robo-advisors). Moreover, research indicates that students are either already using conversational agents (such as chatbots) or are open towards using them (Flanagin 2005; Jain et al. 2018). Thus, students seem to be an adequate sample for our study. The required sample size was determined a priori with GPower (Faul et al. 2007). We assumed an effect size of d = .25, alpha = .05, and power = 0.8 for our study and the power analysis indicated a minimum required sample size of 159 participants for the three treatment groups. Because of potential technical issues or individual mistakes by the participants during the experiment, we invited 200 participants using hroot (Bock et al. 2014). Overall 195 participants (91 females, 104 males, mean age=23.734, SD=4.319) took part in our study. They received a performance-based payment ranging from 2.00 to 4.00 Euro based on their investment and an additional show up fee of 3.00 Euro for participating in the experiment. The mean payoff was 5.686 Euro (SD=0.822) and the whole experiment took approximately 25 minutes (mean=21.435, SD=4.636 minutes). We invited no additional participants after initial data analysis.

After the experiment and before the data analysis, we cleaned the dataset as outlined in the following. First, we removed all participants that made no investment decision, failed to provide all information during the interaction with the RAC, or did not follow the experimental task by analyzing the conversation logs (8 participants). Second, we removed all participants that failed both attention check questions (4 participants). The final dataset includes 183 participants (89 females, 94 males, mean age=23.765, SD=4.366), which are nearly evenly distributed among the three experimental treatment groups. We found no significant differences between the three groups for gender, age, and experience with chatbots. Moreover, except for five participants, none had experience with financial robo-advisory.

4.2 Treatment Structure

Our laboratory experiment applied a between-subjects design. We are interested if a varying degree of anthropomorphizing the RAC influences the users' investment behavior and perception of the RAC. Following existing studies (Go and Sundar 2019; Gong 2008; Seeger et al. 2018; Wünderlich and Paluch 2017), we decided to assess three levels of anthropomorphizing the RAC (LOW, MED, and HIGH). To implement these three levels, we decided to vary certain social cues (Feine et al. 2019) of the RAC in order to be more human-like or less human-like. Following Table 1 depicts the three levels of anthropomorphism used in our experiment with their respective social cue design decision and studies investigating this social cue.

Social Cue	**Level of Anthropomorphism**			**References**
	LOW	**MED**	**HIGH**	
Name	*<NONE>*	"Robo-Advisor"	"Charles"	(Hodge et al. 2018)
Avatar				(Wünderlich and Paluch 2017)
Response time	Instant	Instant	Dynamic	(Appel et al. 2012; Gnewuch et al. 2018a)
Typing Indicator	×	✓	✓	(Gnewuch et al. 2018b)
Greeting & Farewell	×	✓	✓	(Leite et al. 2013; Sabelli et al. 2011)
Self-reference	×	✓	✓	(Nass et al. 1994)
Civility / Thanking	×	✓	✓	(Fogg and Nass 1997)
Remember user's name	×	✓	✓	(Richards and Bransky 2014)

Table 1. Experiment Treatment - Chatbot Anthropomorphic Design

The three RAC in our study varied by the name they displayed during the interaction with the user. In the LOW group, we provided no name, in the MED and HIGH groups we displayed a name for the RAC. In the MED group, we chose the name "Robo-Advisor" and following Hodge et al. (2018), we named the RAC of the HIGH group "Charles". We adapted three avatar pictures investigated by Wünderlich and Paluch (2017) for our three levels of anthropomorphism. In order to simulate a more human-like messaging experience, we delayed the response time of the HIGH group dynamically to the messages send (Appel et al. 2012). In the LOW and MED group, we decided to not delay the RACs' response time. Moreover, the RAC in the HIGH group provides a graphical indication that the message is prepared (i.e. during the dynamic delay), while LOW and MED provided no such graphical indication (Gnewuch et al. 2018b). In addition to these social cues addressing the visual appearance and chronometric behavior, we also decided to vary the way the RAC is texting. In the MED and the HIGH group the RACs greet the participants and provide a farewell (Leite et al. 2013; Sabelli et al. 2011) to simulate human-human communication. In contrast, the RAC of the LOW group does not provide a greeting or farewell message. Similarly, the RACs of the MED and HIGH groups acknowledge input by the participant with a "thank you" (Derrick and Ligon 2014; Fogg and Nass 1997). In addition, the RACs in the MED and HIGH groups refer to themselves in the conversation (Nass et al. 1994) by using personal pronouns such as "I" and "we". In comparison, the RAC in the LOW group avoided this by using passive formulations. Moreover, the RACs in the MED and HIGH groups remembered the participant's name during the interaction and utilized the name when addressing the participant, e.g. for requesting an action from the participant (Richards and Bransky 2014). Summed up, we have three distinct RAC designs that vary in their degree of anthropomorphism based on the dedicated implementation of various social cues.

4.3 Procedure and Scenario

Our experiment consisted of four steps. Upon arrival in our laboratory, the participants signed the consent forms and were seated randomly in the experimental cabins. The entire experiment was performed using the web-based questionnaire LimeSurvey. The experiment started with a general introduction text about the experiment followed by the description of the scenario. Moreover, we randomly assigned the participant into one of the three experimental treatment groups.

After completing the introductory part, one of the three RAC (depending on the participant's group assignment) was included in LimeSurvey and the participants started their interaction with the RAC. The RAC created the user profile by asking questions regarding participant's name, age, and gender as well as elicited their individual risk level using the multiple price list method (Holt and Laury 2005). Subsequently, the actual investment decision task was started. In our experiment, we simulated an online investment decision based on the interaction with a RAC that provides an explanation of the investment options (three fictive companies) and recommends one company. After a short introduction from the RAC, the participants are presented fact sheets on three firms in the semi-inductor industry. Adapted from Hodge et al. (2018), they contain financial information for the recent quarter, a dividend summary, and an industry outlook. The key financial figures for each firm are mere multiples of the other firms' figures, the outlook is the same for the three companies and none of the companies pay dividends. This makes it impossible for the participant to identify a clear favorite investment option, since no firm appears objectively better than the other (Hodge et al. 2018). Furthermore, we randomized the presentation sequence of the fact sheets to avoid order effects. Participants must actively confirm that they thoroughly read the company fact sheets. At the end of this first interaction with the RAC, the participants were asked to invest 3000 MU into one of the three fictive companies. All money related values in the experiment were given in "Monetary Units (MU)" and 100 MU are equal 10 Eurocent.

Next, the participants processed the post-experimental questionnaire with the measures outlined in the next section. After completing the questionnaire, the participants once more interacted with the RAC and the result of their investment was presented. The actual performance of their investment was randomly selected between four outcomes (-1000 MU, -500 MU, +500 MU, +1000 MU).

4.4 Measures

All measures used in the questionnaire were adopted from existing research. Table 2 summarizes the measures, items, sources, descriptive values as well as the factor loadings for each item. To account for alternative explanations of the investigated effects, we additionally included the following control variables: age, gender, prior experiences with chatbots, disposition to trust in technology, prior experiences with robo-advisors, and individual risk level (Holt and Laury 2005; McKnight et al. 2011). The patterns of results remained qualitatively unchanged when including these control variables in our models except for gender and disposition to trust in technology. Accordingly, we will omit all controls except for these two when presenting our results in the subsequent sections. Moreover, 178 out of 183 participants reported no prior experience with robo-advisory, thus, we did not include this factor in our analysis.

Measures	Item	Mean	SD	Loadings
Anthropomorphism (α=.793; M=2.413; SD=1.320) (Epley et al. 2007)				
The robo-advisory chatbot has intentions.	*ANP1*	4.093	1.845	*.494 (dropped)*
The robo-advisory chatbot can experience emotion.	*ANP2*	2.055	1.421	*.567 (dropped)*
The robo-advisory chatbot has a free will.	*ANP3*	1.907	1.312	.789
The robo-advisory chatbot has consciousness.	*ANP4*	2.383	1.599	.815
The robo-advisory chatbot has a mind of its own.	*ANP5*	2.951	1.764	.740
Social Presence (α=.864; M=3.443; SD=1.398) (Gefen and Straub 2003; Qiu and Benbasat 2009)				
There is a sense of human contact in the robo-advisory chatbot.	SP1	4.410	1.597	0.794
There is a sense of personalness in the robo-advisory chatbot.	SP2	2.754	1.569	*0.429 (dropped)*
There is a sense of sociability in the robo-advisory chatbot.	SP3	3.437	1.682	0.796
There is a sense of human warmth in the robo-advisory chatbot.	SP4	2.82	1.659	0.744
There is a sense of human sensitivity in the robo-advisory chatbot.	SP5	3.104	1.698	0.728
Trusting Beliefs (α=.914; M=4.949; SD=1.029) (McKnight et al. 2002; Qiu and Benbasat 2009; Wang and Benbasat 2005)				
I believe that the robo-advisory chatbot would act in my best interest.	TB1	4.874	1.537	0.797
If I required help, the robo-advisory chatbot would do its best to help me.	TB2	5.033	1.410	0.726
The robo-advisory chatbot is interested in my well-being, not just its own.	TB3	4.503	1.596	0.693
The robo-advisory chatbot is truthful in its dealings with me.	TI1	4.847	1.366	0.822
I would characterize the robo-advisory chatbot as honest.	TI2	4.902	1.494	0.806
The robo-advisory chatbot would keep its commitments.	TI3	5.814	1.063	0.623
The robo-advisory chatbot is sincere and genuine.	TI4	4.546	1.341	0.725
The robo-advisory chatbot is competent and effective in providing financial advice.	TC1	5.104	1.397	0.707
The robo-advisory chatbot performs its role of giving financial advice very well.	TC2	5.055	1.386	0.731
Overall, the robo-advisory chatbot is a capable and proficient financial advice provider.	TC3	4.809	1.302	0.708
In general, the robo-advisory chatbot is very knowledgeable about finance.	TC4	5.098	1.322	0.599
Disposition to Trust in Technology (α=.868; M=4.499; SD=1.370) (McKnight et al. 2011)				
My typical approach is to trust new IT until they prove to me that I shouldn't trust them.	TIT1	4.634	1.552	0.918
I usually trust in information technology until it gives me a reason not to.	*TIT2*	4.65	1.554	0.918
I generally give an information technology the benefit of the doubt when I first use it	TIT3	4.213	1.513	0.782
α = Cronbach's alpha | M = Mean | SD = standard deviation				

Table 2: Measures used in the post-experimental questionnaire

To test whether our manipulation of the RAC with respect to the three levels of anthropomorphism was successful, we assessed users' perceived anthropomorphism of the RAC (Epley et al. 2007). A one-way ANOVA revealed a significant effect of the treatment condition on perceived anthropomorphism (F(2,180)=3.579, p=.030). The results of a Tukey HSD post-hoc comparison showed that there was a significant difference between the LOW condition (M=2.44, SD=1.05) and the HIGH condition (M=3.00, SD=1.28). However, the differences between the MED condition (M=2.73, SD=1.28) and both the LOW and HIGH conditions were not significant. Despite these insignificant results, we did not exclude the MED condition because its mean was in the middle of the range between LOW and HIGH and we believe that with a larger sample one should be able to also observe significant differences for the MED condition.

5 Experiment Results

To test our proposed hypotheses, we conducted the following two steps. First, we conducted a one-way ANCOVA to test the effect of the three levels of anthropomorphism on the users' perceived social presence of the RAC (H1). Second, we estimated three regression models to examine the relationships between perceived social presence, trusting beliefs, and users' likeliness to follow the recommendation by the RAC (H2, H3, H4). All regression analyses were performed in Stata version 13.0. Table 3 summarizes the descriptive results and Table 4 displays the results of the regression analyses.

Group	N	Gender[1]	Chatbot Experience [2]	Perceived anthropomorphism [3]	Perceived social presence [3]	Trusting beliefs [3]	Disposition to trust technology [3]	Individual risk level [4]	Followed RAC recommendation
LOW	62	(30 | 32)	(29 | 33)	2.442 (1.047)	2.806 (1.186)	4.811 (1.058)	4.505 (1.291)	5.323 (2.125)	83.87 %
MED	61	(29 | 32)	(31 | 30)	2.731 (1.103)	3.459 (1.387)	5.125 (1.002)	4.694 (1.341)	5.180 (1.866)	85.25 %
HIGH	60	(30 | 30)	(22 | 38)	3.000 (1.277)	4.083 (1.333)	4.953 (0.985)	4.294 (1.468)	6.217 (2.043)	81.67 %
[1] (female | male) users | [2] (novice | experienced) chatbot users | [3] measured on a 7-point Likert scale |[4] measured on a scale from 1 to 10									

Table 3. Descriptive Results

First, consistent with H1, the results of a one-way ANCOVA confirmed a significant effect of the levels of anthropomorphism on perceived social presence ($F(2,178)=15.37$, $p<.001$). Thereby, we controlled for gender and user's disposition to trust in technology. Post-hoc Tukey HSD comparisons revealed significant differences between the LOW condition ($M=2.81$, $SD=1.19$) and the MED condition ($M=3.46$, $SD=1.39$) as well as between the LOW condition and the HIGH condition ($M=4.08$, $SD=1.33$). Further, there was a significant difference between the MED condition and HIGH condition. Additionally, gender and disposition to trust in technology were significant control variables. More specifically, we found that females ($M=3.70$, $SD=1.36$) yield significantly higher levels of perceived social presence than males ($M=3.20$, $SD=1.40$). Taken together, these results confirm the proposed positive effect of anthropomorphizing a RAC on users' perceptions of the RAC's social presence, thus supporting H1.

To test H2, we ran a linear regression (regression I) to assess the effect of perceived social presence on users' trusting beliefs in the RAC. The results indicate that social presence significantly influences users' trusting beliefs ($\beta=0.217$, $p<.001$). Furthermore, we found that female users ($\beta=0.324$, $p=.031$) yield a higher level of trusting beliefs and that users' disposition to trust in technology ($\beta=0.147$, $p=.012$) significantly influences their trusting beliefs. Therefore, these results confirm the proposed positive effect of social presence on trusting beliefs, thus supporting H2.

Finally, we assessed the effect of social presence and trusting beliefs on the users' likeliness to follow the recommendation given by the RAC (H3 and H4). First, we conducted a binary logistic regression (regression II) to test the effect of perceived social presence on users' likeliness to follow the recommendation. The results indicate that there is no significant effect of social presence on likeliness to follow ($\beta=-0.171$, $p=.256$), thus rejecting H3. Second, to test H4, we estimated another binary logistic regression (regression III) assessing the effect of trusting beliefs and social presence on the likeliness to follow the recommendation by the RAC. Consistent with H4, the analysis reveals that trusting beliefs significantly influences likeliness to follow the recommendation by the RAC ($\beta=0.601$, $p=.003$, OR=1.824 [95% CI: 1.219, 2.729]). Moreover, when including trusting beliefs, the effect of social presence on likeliness to follow approached significance ($\beta=-0.313$, $p=.052$, OR=0.731 [95% CI: 0.533, 1.003]).

Dependent variable	(I) Trusting beliefs	(II) Likeliness to follow	(III) Likeliness to follow
Intercept	3.382*** (0.296)	1.199 (0.905)	-0.712 (1.153)
Factors			
Social presence	0.217*** (0.0607)	-0.171 (0.150)	-0.313+ (0.161)
Trusting beliefs			0.601** (0.206)
Controls			
Gender: female	0.324* (0.149)	0.0976 (0.411)	-0.113 (0.412)
Disposition to trust in technology	0.147* (0.0580)	0.225 (0.163)	0.143 (0.167)
F / χ^2	F(3, 179)=12.222***	$\chi^2(3)=2.81$	$\chi^2(5)=12.13$*
Observations	183	183	183
R^2	$R^2 = 0.170$	R^2Pseudo = .020	R^2Pseudo = .066
*Note: + < 0.1; * < .05; ** < 0.01; *** < 0.001 | Robust standard errors in parentheses*			

Table 4. Regression analyses results

Given the results as described above, we conducted a mediation analysis with 10,000 bootstrap samples (Hayes 2018, Model 4) to test whether trusting beliefs mediate the relationship between perceived social presence and likeliness to follow the recommendation by the RAC. We defined perceived social presence as the independent variable, trusting beliefs as the mediator, and likeliness to follow the recommendation as the dependent variable. The indirect effect of social presence on likeliness to follow the recommendation was statistically significant (β=0.158, SE=0.062, [95% CI: 0.054, 0.2970]), indicating that trusting beliefs mediates the relationship between social presence and likeliness to follow the recommendation by the RAC. The direct effect of social presence on likeliness to follow the recommendation was not significant (β=-0.310, p=.051). The effects of social presence on trusting beliefs (β=0.252, p<.001) and trusting beliefs on likeliness to follow the recommendation by the RAC (β=0.627, p=.004) were still significant. Therefore, these results show that social presence has a positive indirect effect on likeliness to follow the recommendation via trusting beliefs.

6 Discussion

Our study investigates how anthropomorphism in the design of an RAC influences users' perceived social presence, trusting beliefs, and ultimately, their likeliness to follow the recommendation by the RAC. We found support for hypotheses H1 and H2 confirming the proposed effects of an increasing level of anthropomorphism on the participants' perceived social presence and trusting beliefs. Our findings are in line with existing research (e.g., Araujo 2018; Go and Sundar 2019; Hess et al. 2009; Qiu and Benbasat 2009) and add to the body of knowledge on how to design chatbots yielding a high level of social presence and trust for certain contexts (in our case: investment robo-advisory). In our experiment, the participants were asked to decide between three fictive investments and the RAC recommended one investment. There was no significant difference in the likeliness to follow the recommendation by the RAC between the three experimental groups ($\chi^2(2)$= 0.288, p=.866). Perceived social presence did not directly impact users' likeliness to follow the recommendation (H3), but there was a positive indirect effect on likeliness to follow the recommendation via trusting beliefs. Moreover, our results suggest that users' trusting beliefs had a significant impact on their likeliness to follow the recommendation by the RAC (H4).

6.1 Theoretical Contributions

Our first contribution is to show that the RAC's anthropomorphic design not only influences users' perceptions (i.e., social presence, trust) but also their behavior (i.e., whether or not they follow the recommendation by the RAC). Therefore, when offering robo-advisory in the form of a chatbot, one important aspect to be considered is to design the RAC in such a way that the users' trust and follow the investment advices provided by the RAC (Hodge et al. 2018; Jung, Dorner, Weinhardt, et al. 2018).

Our second contribution is to highlight the mixed effects of gender on social presence and trusting beliefs. Female participants yielded a higher level of both measures. This is in line with recent research highlighting the importance to consider users' gender in order to achieve the intended level of trust (Beldad et al. 2016) and credibility (Nowak and Rauh 2005). Besides, the increased level of perceived social presence for female participants is consistent with findings from Thayalan et al. (2012) who found females to have a higher perception of social presence in an e-learning environment.

Our third contribution is to provide a more nuanced understanding of how anthropomorphism affects users' likeliness to follow the RAC's advice. The lack of a direct effect of the anthropomorphism on the participants' likeliness to follow, but significant effect of their trusting beliefs in the chatbot, can be explained with the CASA paradigm and social response theory (Nass and Moon 2000). During the interaction with RAC, the participants experience various social cues instantiated in the RAC and these cues trigger a mindless behavior by the participants resulting in a behavioral response based on their prior experiences and expectations (Nass and Moon 2000). The different implementations of the social cues influence the participant's perceived social presence of the RAC, which subsequently also has a positive effect on their trusting beliefs towards the RAC. While trusting beliefs are a significant predictor of users' investment decision-making (i.e., follow the recommendation by the RAC or not), we find no direct or mediated effect of perceived social presence on likeliness to follow the recommendation by the RAC. This finding could indicate a potential Uncanny Valley effect (Mori et al. 2012) in our study. Our results suggest that simply increasing social presence to the maximum is not reasonable in this context. On the contrary, designers need to consider to what extent social presence should be increased in order to achieve a high level of trusting beliefs. In addition, it is important to investigate additional factors that influence users' trusting beliefs so that the design of as well as the interaction with RAC can be further enhanced to increase the likeliness that users accept investment recommendations by RAC.

6.2 Limitations and Future Research

There are several limitations to this study that should be considered when assessing our findings. In the following, we discuss these limitations and outline avenues for future research. First, our experimental design considered only three levels of anthropomorphism in the design of the RAC, which were manifested through the design of eight social cues. While there is empirical evidence for each of the implemented social cues to affect users' perceived anthropomorphism (see Table 1), the structure of each level was not derived from an established framework. Therefore, we assumed that the combination of these cues represents an adequate manifestation of three distinct levels of anthropomorphic design. However, as the results of our manipulation check suggest, it is not clear to what extent which social cues individually or in combination affect user' perceived anthropomorphism. Therefore, it is a limitation of our study that we did not find signficant differences in perceived anthropomorphism between the medium and low/high level of anthropomorphic design of the RAC. Moreover, the construct anthropomorphism seems to measure aspects that go beyond what we considered in the RAC's design, such as the intelligence and a free will of the RAC. Therefore, other measures such as perceived humanness (Holtgraves et al. 2007), could be more appropriate in our context and therefore, could be verified in future research. Moreover, our study lacks a baseline in terms of social presence and anthropomorphism to assess the effect of using a CUI for robo-advisory. While the conversational character of the interaction itself can be understood as an anthropomorphic social cue (McTear 2002; Nass et al. 1994), we had no experimental treatment group with an absolutely non-anthropomorphic design (i.e. no anthropomorphic cues implemented). Commercial robo-advisors are only implemented in form of GUIs, thus, a treatment group that passes through the same advisory process using a GUI only without a chatbot could have led to other or stronger differences between treatments. However, our study focuses on the interaction of users with robo-advisory using a CUI. This decision has been taken because we intended to utilize the proposed positive effects of using natural language for the interaction on users' perceived social presence and trusting beliefs following the CASA paradigm and social response theory (Nass and Moon 2000). Future research could, for example, investigate the effects of combining a CUI and GUI for providing robo-advisory.

Second, although our experimental task was based on a realistic robo-advisory process, it had a considerably lower complexity. The investment decision (i.e., choosing from three options and deciding to follow the recommendation by the RAC or not) was rather simple in contrast to more complex financial decision-making in reality. Participants' expectation to gain more monetary payout has been found to be an appropriate motivation in an experimental context, but more research is required to understand the effect of a CUI (e.g. in the form of a text-based chatbot or a voice-based conversational agent) in a real-life financial context. In our experiment, there were no costs associated with consulting the RAC, the decision to invest in one of the companies, and ultimately (not) following the recommendation by the RAC. Therefore, we overserved a relatively high likeliness to follow the RAC's advice (82% – 85%) among our three experimental groups. In contrast, existing robo-advisor services require users to pay about 0.5% – 2% of their investment sum per year. Moreover, because of the relatively low complexity of the experimental task, the interaction time with the RAC was rather short (about 5 minutes on average). Therefore, we examine only participants' first impressions of the RAC. Future research could increase the complexity and realism by adding several iterations in the experimental process to include the rebalancing phase (Jung, Dorner, Weinhardt, et al. 2018). Several iterations would also better reflect long-term usage, which has not been investigated in this study, but could potentially lead to different results and/or to a richer explanation of the perceptions and usage of RACs.

Third, our robo-advisory process included a risk assessment based on the multiple pricelist method (Holt and Laury 2005) and we found no significant effect of the participants' risk-level in our analysis. Although we assumed an equal distribution of the participants' risk level between our three groups, we identified significant differences between the three treatment groups in a subsequent ANOVA test (F(2, 180)=4.698, p=.010). The TukeyHSD post-hoc analysis reveals that the HIGH (mean=6.217, SD=2.043) group has a significantly higher risk level than the LOW (mean=5.323, SD=2.125) and MED (mean=5.180, SD=1.866) groups. Furthermore, the risk-assessments implication on the investment was not clearly stated during the experimental process, since the assessment only served to assess the participants' risk level and to increase participants' immersion in the investment advisory scenario. Because we lack a baseline risk level (i.e. risk level assessment before the interaction with the RAC), a further investigation of this interesting finding is subject to future research.

Fourth, there might be other theoretical explanations of the observed effects, such as the uncanny valley assumption (Mori et al. 2012). According to this assumption, an overly human-like or anthropomorphic design of the RAC might have triggered unintended negative effects, such as a reduced level of trust. Future research could, for example, investigate whether there is a certain, good enough level of anthropormophism that yields a maximum level of trust into the RAC, while not being perceived as uncanny because of a too high level of anthropormorphism.

Fifth, our student sample was selected purposefully in order to have a homogeneous group of participants with low or no financial expertise. Future studies could employ participants with a higher level of expertise to assess our findings as well as to investigate the potential effect of finanical expertise on the users' perceptions of the RAC and investment decisions.

Summed up, we outlined three potential avenues for further research, but there is more research required in order to understand the interaction between users and robo-advisory in the form of chatbots as well as how the design of certain social cues of RAC affect this interaction.

7 Conclusion

In this paper, we present our study investigating the effects of anthropomorphism on users' perceptions of an RAC and their likeliness to follow its investment recommendations. To summarize, we provide valuable empirical knowledge to the increasing knowledge base on the interaction between users and robo-advisors. Our findings have implications for the design of RACs and we can make several recommendations for practice and future research. First, it is imperative to ensure a high level of trust towards the offered robo-advisory in general and in the form of a chatbot in particular. Second, in the context of chatbots, the results of our study as well as existing research found that an increased level of social presence of the agent is positively correlated with the users' trusting beliefs. Thus, it is recommended to ensure an adequate level of perceived social presence, for example by an appropriate social cues design of RAC. Furthermore, we found different results for the female participants (i.e. a higher level of social presence and trusting beliefs), which suggest that when designing a RAC the potential users' gender as well as other user characteristics (e.g. disposition to trust in technology) should be considered.

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